GRE Counting Basics Cheat Sheet

1. Factorial (!)

$$n! = n \cdot (n-1) \cdot (n-2) \cdots 2 \cdot 1$$

Example:

$$4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$$

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2. Permutation (Order Matters)

Formula:

$$nP_r = \frac{n!}{(n-r)!}$$

Arrange r items from n with order.
Example: 3 people (A, B, C) in 2 seats:

$$3P_2 = \frac{3!}{(3-2)!} = 6$$

Arrangements: AB, BA, AC, CA, BC, CB

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3. Combination (Order Doesn't Matter)

Formula:

$$nC_r = \frac{n!}{r! \cdot (n-r)!}$$

Choose r items from n without caring about order.
Example: Choose 2 from 3 (A, B, C)

$$3C_2 = \frac{3!}{2! \cdot 1!} = 3$$

Teams: AB, AC, BC

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4. Quick Tip to Decide

Question	**Use**
Does order matter?	Permutation
Does order NOT matter?	Combination

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5. Common GRE Problems & Patterns

Problem Type	Concept	Formula
Arrangements of objects	Permutation	$nP_r = \dfrac{n!}{(n-r)!}$
Selecting team members / groups	Combination	$nC_r = \dfrac{n!}{r!(n-r)!}$
Repetition allowed (Permutation)	n^r	
Repetition allowed (Combination)	$\binom{n+r-1}{r}$	

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6. Examples to Memorize

Problem	Solution	Type
Arrange 4 books on a shelf	$4! = 24$	Permutation
Choose 2 books from 5	$5C_2 = 10$	Combination
3-digit PIN with digits 0–9	$10^3 = 1000$	Permutation with repetition

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Summary

- **Factorial** = total number of ways to arrange all items.
- **Permutation** = arrangement when order matters.
- **Combination** = selection when order doesn't matter.
- **With repetition** = use powers or special formulas.